Advertising Performance Analytics

SQL, Excel, and Power BI Analysis

Prepared by: Rishabh Shindore
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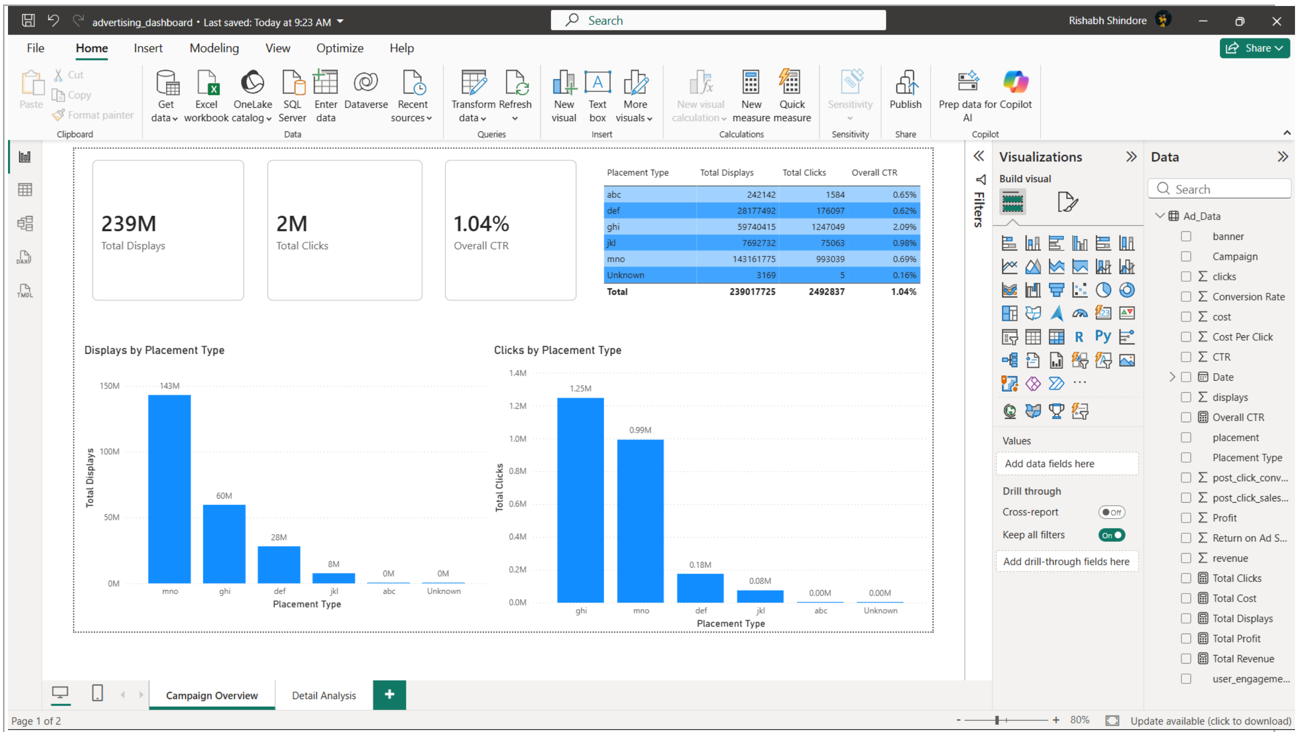
1. Executive Summary

This project analyzes advertising performance across campaigns and placement types using SQL, Excel, and Power BI. The analysis consolidates displays, clicks, cost, revenue, profit, placement, campaign, date, banner, and user-engagement information into a validated reporting workflow.

The final Power BI report provides KPI cards, placement-level charts, a summary table, and interactive filters for campaign, date, placement type, banner, and user engagement.



Power BI overview.

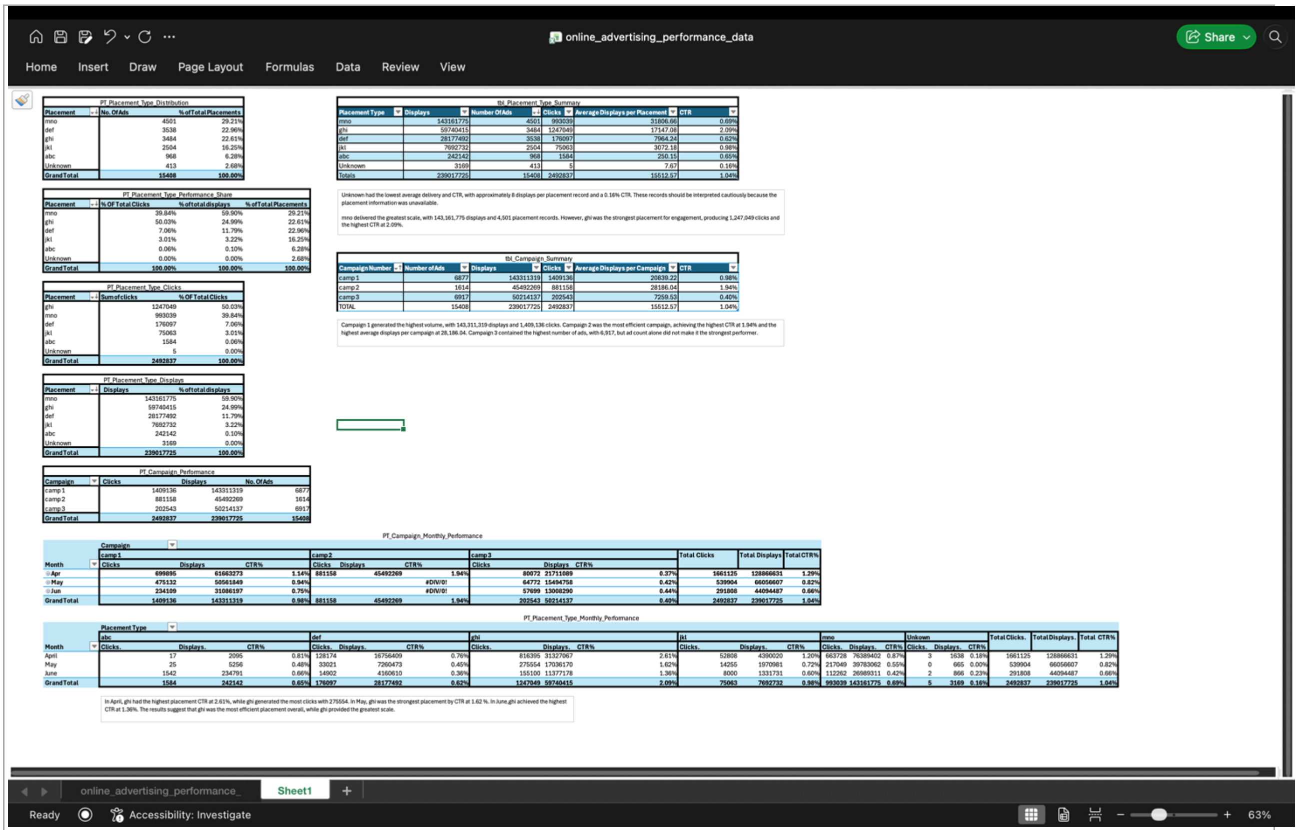
2. Project Objective

The objective was to create a reliable and interactive advertising-performance analysis that answers how campaigns and placements performed, how much exposure and engagement they generated, and whether the final dashboard reconciles with the SQL and Excel results.

3. Dataset Overview

The dataset contains 15,408 advertising records. Key measures include displays, clicks, cost, revenue, and profit. Descriptive fields include date, campaign, banner, placement, cleaned placement type, and user engagement.

The analysis uses the raw data as the primary source. Summary and PivotTable outputs were used for comparison and validation rather than as the main Power BI source.

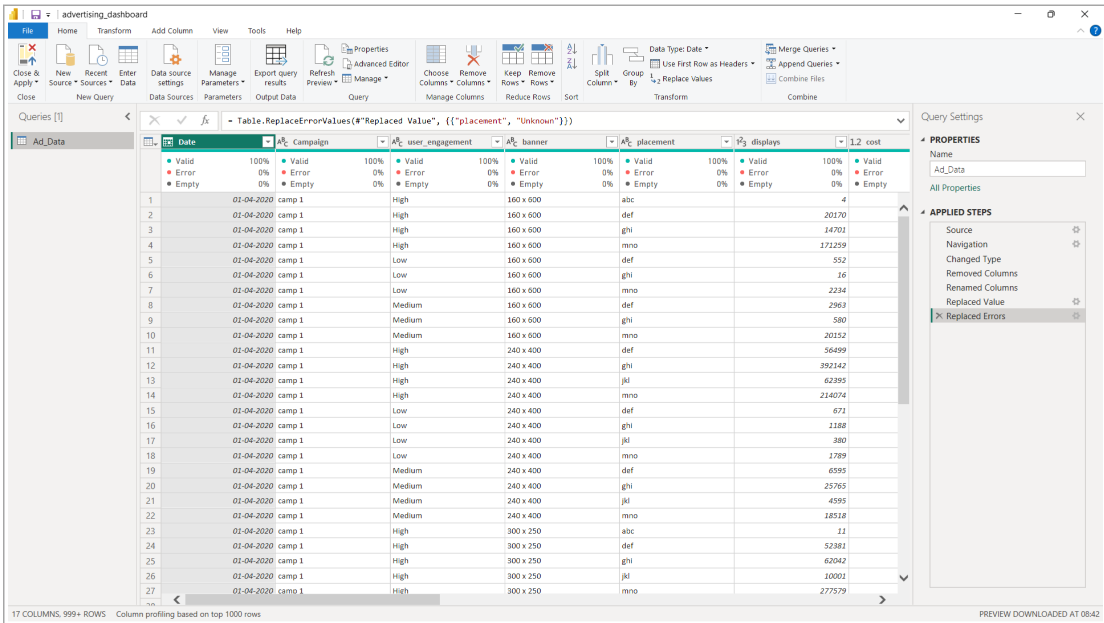


Excel Data Sheet Screenshot

4. Data Preparation and Quality Checks

Data was reviewed for field names, data types, blank columns, duplicate calendar fields, and missing placement values. The Date field was retained as the primary date field, while redundant day and month fields were removed where they duplicated information available from Date.

The 413 records with missing placement values were retained and labeled Unknown so their valid performance measures were not excluded from the analysis. Unused blank columns were removed where applicable.

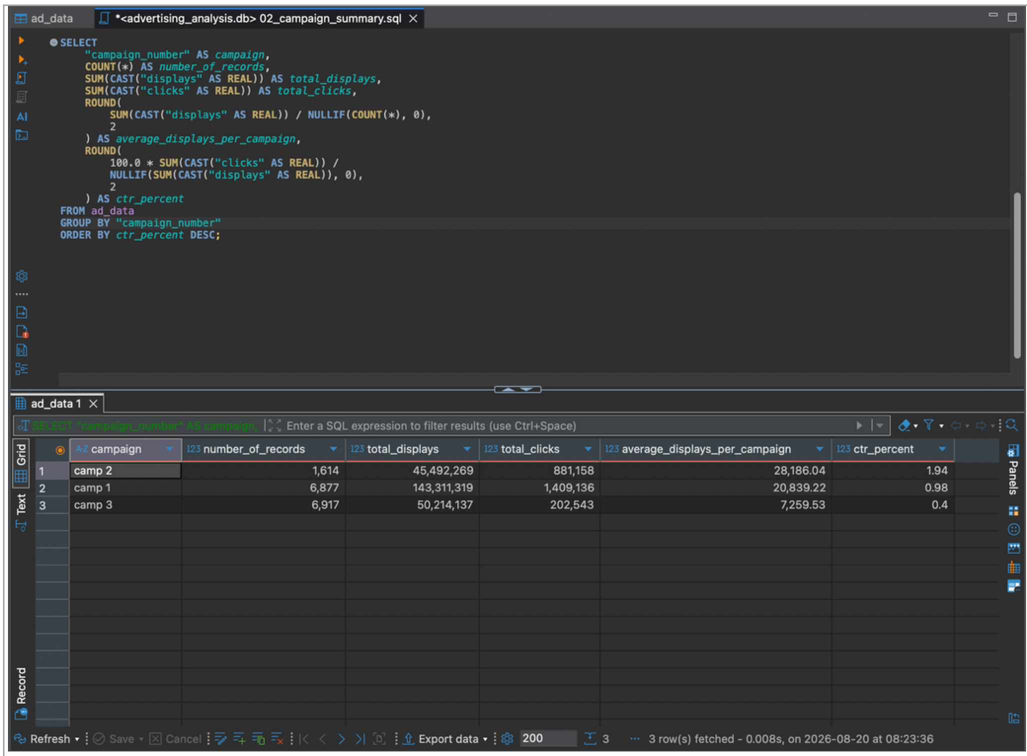


Power Query screenshot.

5. SQL Analysis

SQL queries were used to validate the row count, inspect the source fields, identify placement categories, summarize placement and campaign performance, review monthly performance, and calculate overall totals and CTR.

The main KPI calculations use total clicks divided by total displays. The SQL scripts are included in the project SQL folder and provide a reproducible audit trail for the analysis.

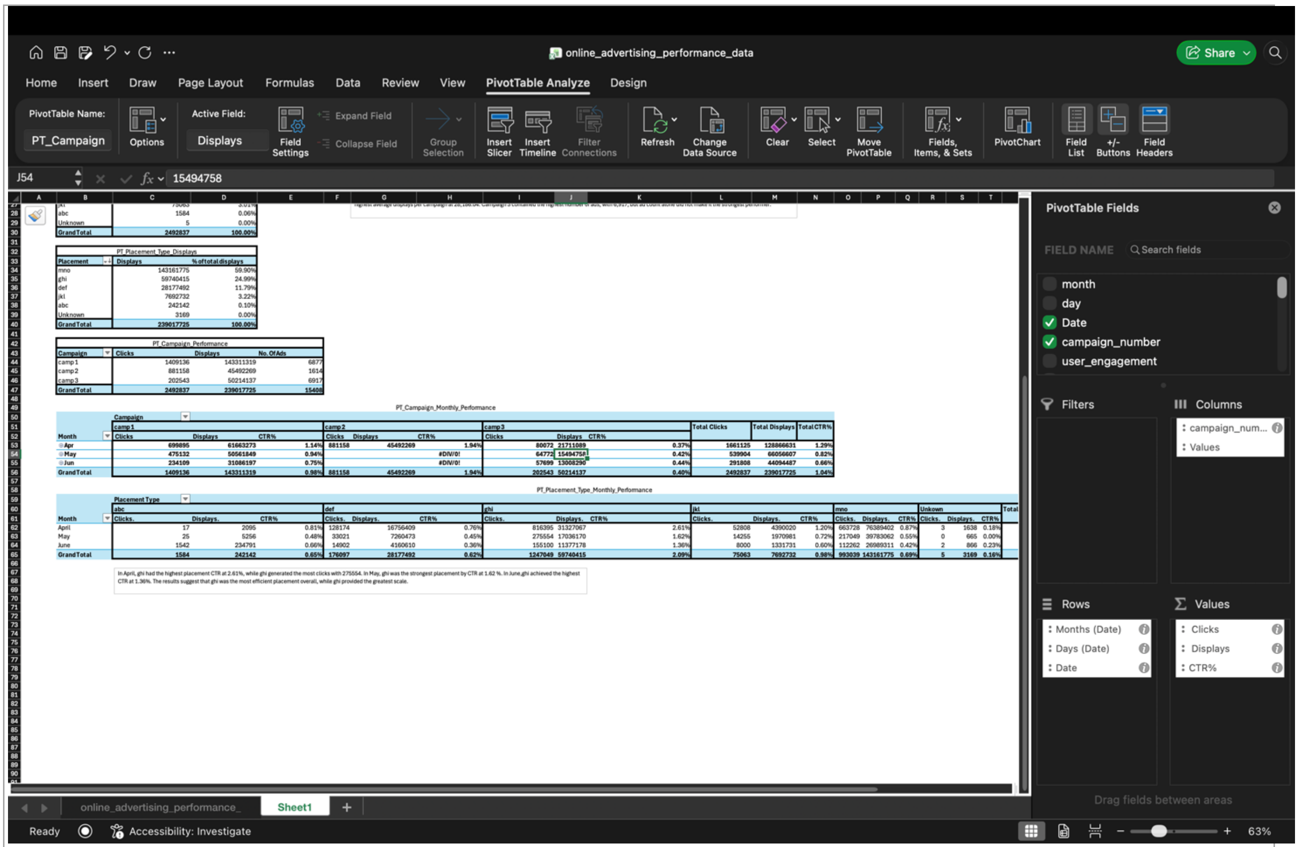


SQL query-result screenshot.

6. Excel Analysis

Excel PivotTables were used as an intermediate analytical and reconciliation layer. They summarized performance by placement, campaign, and time period and provided a reference point for validating the SQL and Power BI outputs.

The Excel workbook is included in the project Data folder as supporting evidence.

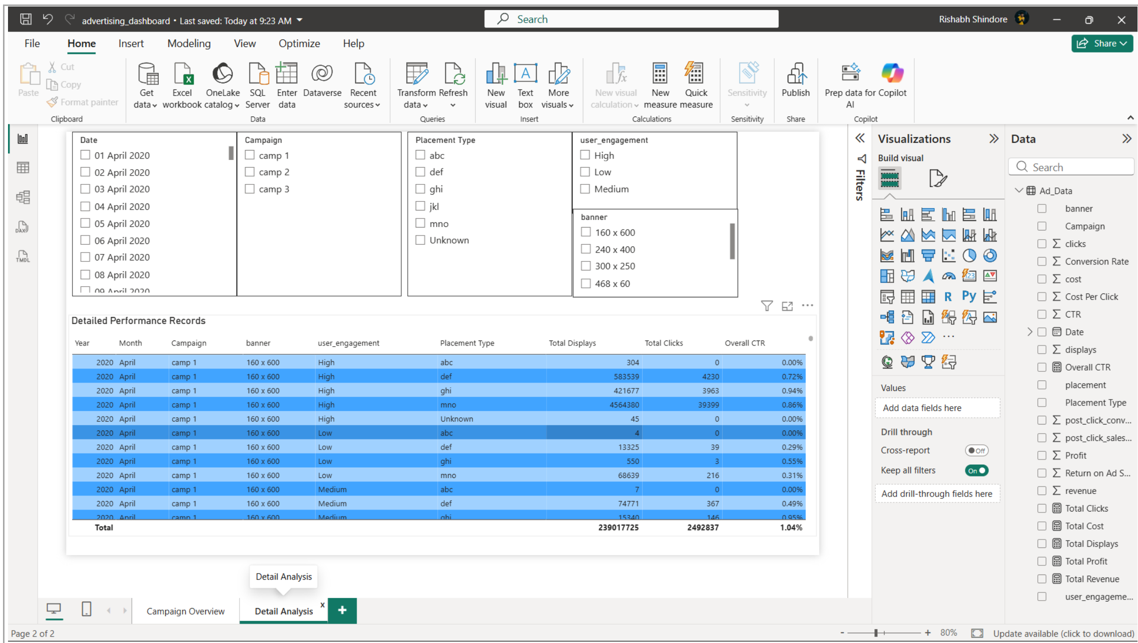


Excel PivotTable screenshot here.

7. Power BI Dashboard

The Power BI report contains an overview page and a detail-analysis page. The overview page includes cards for total displays, total clicks, and overall CTR, together with placement charts and a placement summary table.

The detail page includes slicers for date, campaign, placement type, banner, and user engagement. The main slicers were synchronized across report pages to support interactive exploration.



Power BI overview and detail-page screenshots.

8. Key Findings

The final validated totals are shown below. These figures represent the complete dataset with no slicer restrictions applied.

Advertising Performance Analytics

Project Notes and Key Findings

This section documents the dataset, data-quality treatment, placement performance, campaign performance, and validated totals for inclusion in the analytics report

Dataset and Data Quality

The dataset contains **15,408 records** across six cleaned placement categories: **abc, def, ghi, jkl, mno,** and **Unknown**.

The placement field contains **413 text values labelled #N/A**, representing unavailable placement information. These values were retained and categorised as **Unknown** for analysis so that their associated performance measures were not excluded.

Validated dataset totals

Metric	Result
Total records	15,408
Total displays	239,017,725
Total clicks	2,492,837
Overall CTR	1.04%
Unknown placement records	413

Placement Representation and Delivery

The **mno** placement had the highest representation, accounting for **4,501 records**, or **29.21%** of the dataset.

mno also received the highest number of ads served, with **143,161,775 displays**, representing **59.90%** of all displays.

Display volume was therefore highly concentrated in mno. Although mno represented only 29.21% of placement records, it accounted for 59.90% of total displays. By contrast, def, jkl, and abc represented larger shares of records than displays, indicating lower display volume per record.

Average displays per placement record

Placement	Total displays	Records	Average displays per record
mno	143,161,775	4,501	31,807
ghi	59,740,415	3,538	16,885

Unknown	3,169	413	8
Other categories	See SQL/Power BI output	—	Lower than mno and ghi

Interpretation

Ad delivery was strongly concentrated in mno. This placement was the primary high-volume placement, while abc and Unknown contributed very little display volume per record. Unknown should be interpreted cautiously because its placement information was unavailable.

Clicks and Engagement

The **ghi** placement generated the highest number of clicks, with **1,247,049 clicks**, representing **50.03%** of all clicks.

ghi generated 50.03% of all clicks from 24.99% of displays, while mno generated 39.84% of clicks from 59.90% of displays. This indicates stronger click concentration in ghi and higher display volume but relatively lower click contribution in mno.

Placement CTR ranking

Rank	Placement	CTR
1	ghi	2.09%
2	jkl	0.98%
3	mno	0.69%
Lowest	Unknown	0.16%

Interpretation

mno delivered the greatest scale, with 143,161,775 displays and 4,501 placement records. However, ghi was the strongest placement for engagement, producing 1,247,049 clicks and the highest CTR at 2.09%. Unknown had the lowest average delivery and CTR, with approximately 8 displays per record and a 0.16% CTR.

Campaign Performance

Campaign 1 generated the highest volume, with **143,311,319 displays** and **1,409,136 clicks**.

Campaign 2 was the most efficient campaign, achieving the highest CTR at **1.94%** and the highest average displays per campaign at **28,186.04**.

Campaign 3 contained the highest number of ads, with **6,917**; however, ad count alone did not make it the strongest performer.

Interpretation

Campaign 1 was the scale leader, while Campaign 2 was the efficiency leader. Campaign 3 had the highest record count but should not be evaluated using volume alone.

Monthly Campaign Performance

Campaign 2 was the strongest campaign in April, achieving the highest observed monthly CTR of **1.94%**. However, Campaign 2 was not active during May or June.

Campaign 1 achieved the highest CTR in both May and June, indicating that it became the leading active campaign after April.

Campaign 3 recorded the lowest observed CTR at **0.37%** in April and should be reviewed for possible optimization.

Recommended interpretation

Campaign performance changed by month. The strongest campaign overall should therefore be interpreted alongside campaign availability and the period in which each campaign was active.

Overall Conclusions

The analysis shows a clear distinction between scale and efficiency. mno was the dominant delivery placement, while ghi generated the strongest engagement response. At campaign level, Campaign 1 generated the greatest volume, whereas Campaign 2 achieved the strongest efficiency when active.

The Unknown category should remain visible in reporting because it contains valid records, but its results should be treated cautiously until placement information can be recovered or improved at source.

Recommendations

Prioritize investigation of ghi to understand the factors contributing to its 2.09% CTR and high click contribution.

Review the role of mno: it provides substantial scale, but its 0.69% CTR suggests an opportunity to improve engagement efficiency.

Audit Campaign 3, particularly its April performance, where the observed CTR was 0.37%.

Improve placement data capture to reduce the 413 Unknown records and strengthen placement-level decision-making.

Monitor both delivery volume and CTR so that high-scale placements are not evaluated on volume alone.

9. Validation Results

The SQL, Excel, and Power BI results were reconciled. The Power BI KPI cards matched the final SQL validation totals, confirming that the imported data and measures were working as intended.

The report was also checked after clearing slicers, and the Unknown placement category remained visible for the 413 records with missing placement.

10. Limitations and Recommendations

The analysis depends on the quality and completeness of the source fields. Missing placement values were handled as Unknown, which preserves the records but limits placement-specific interpretation for those rows.

Future improvements could include a dedicated date table, documented data-refresh procedures, additional campaign profitability measures, and a standardized source-data dictionary.

11. Conclusion

The project produced a complete advertising-performance workflow using SQL for structured analysis, Excel for reconciliation, and Power BI for interactive reporting. The final dashboard is supported by validated totals and documented handling of missing values.

12. Data Source & Disclaimer

This project uses a publicly available advertising dataset downloaded from Kaggle for educational and portfolio purposes. The dataset is not a live production database and should not be interpreted as representing actual business operations, real customer activity, or current advertising performance.

The data may be synthetic, anonymised, simulated, or otherwise prepared for learning and analysis. All findings in this report apply only to the supplied dataset and should not be generalized to a real organization without independent validation.

The dataset was cleaned and transformed for analysis. In particular, 413 records with unavailable placement information were categorized as Unknown. The SQL database, Excel workbook, and Power BI report were created as part of this analytical exercise.

Dataset credit and licensing information should follow the original Kaggle dataset page and its stated license. The original dataset creator and Kaggle source should be acknowledged wherever the dataset is used. Kaggle recommends checking the dataset's license and attribution requirements before reuse.

https://www.kaggle.com/datasets/naniruddhan/online-advertising-digital-marketing-data?resource=download

Dataset Title - Online Advertising Digital Marketing Data

Dataset Creator - N. aniruddhan
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